Relevant Reading

- Engineering Design Optimization
  - Chapter 1: Introduction (32 pages)
- Optimal Loading Document

Problem 1 Vocabulary

Explain the following terms; making sure to use sufficient detail, including any math or helpful figures. In some cases, these terms are simple one sentence definitions, in others, you should include several paragraphs to explain them fully.

Blade Element Momentum Theory

- Tip correction factor

Optimization

- Design Variables
- Objective
- Constraints

Problem 2 Theory

Obtain the optimal loading document from your graduate student mentor, read sections 1, 2.1, and 3. Answer the following questions:

i. What is the axial induction factor of maximum theoretical power? Why is it that value?

ii. How do you calculate the chord and twist of a rotor blade with theoretical maximum power?

iii. What are the weaknesses of this maximal loading approach? What does it not account for? How could you improve it?

Problem 3 Exploration

Complete the following exploration.

i. Use the maximum angle of attack of the maximum angle of attack and implement the optimal loading theory of a rotor blade with theoretical maximum power. Find the chord and twist of the rotor blade. Compare the results to the rotor blade you designed in the previous assignment.